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Multi-Domain Vision Transformer Fusion for Intersectional Demographic Classification from Facial Images

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ABSTRACT Simultaneous recognition of facial demographic attributes such as race and gender faces challenges from expression variations, biological aging, phenotypic overlap, and the limitations of single-domain representations. This study proposes a multi-domain latent feature fusion framework that integrates visual representations from three task-specific pre-trained Vision Transformer (ViT) models for facial biometrics, affective expression, and age estimation, coupled with classical machine learning pipeline optimization for classifying six intersectional demographic subgroups on the DemogPairs dataset. Latent representations are extracted offline from each model to produce a unified visual representation that combines complementary cross-domain information. Seven feature configurations are evaluated through 5-Fold Stratified Cross-Validation across four classifiers: Random Forest (RF), Gaussian Naive Bayes (GNB), Logistic Regression (LR), and Support Vector Machine (SVM) with hyperparameter tuning via Grid Search Cross-Validation (GridSearchCV). Experimental results demonstrate that tri-domain fusion achieves the best performance across three of the four evaluated classifiers, with the SVM model yielding the highest performance, achieving an Accuracy of 93.70%, Precision of 93.72%, Recall of 93.70%, and F1-Score of 93.69% on independent test data. In-depth evaluation across the six subgroups records an F1-Score range between 91.74% and 96.14%, proving the effectiveness of multi-domain latent representation fusion in recognizing intersectional demographic attributes.

INDEX TERMS Race and gender classification, intersectional demographic recognition, Vision Transformer, multi-domain feature fusion, algorithmic fairness.

I. INTRODUCTION

Automated recognition of facial demographic attributes plays a pivotal role in various intelligent application domains, including digital forensic systems, biometric access control, human-computer interaction, and personalized interactive services. However, numerous studies have reported that facial recognition systems frequently exhibit performance disparities across particular demographic subgroups, with substantial accuracy degradation consistently observed in female groups and darker-skinned populations. These accuracy disparities stem from diverse fundamental factors, encompassing imbalanced training data distributions across public datasets and the limitations of conventional feature extractors in capturing visual representations invariant to pose variations and natural visual degradations. The limitations of conventional systems in coping with these visual variations emphasize that fair and accurate demographic recognition still confronts sig-

nificant obstacles, thereby necessitating novel approaches capable of comprehensively modeling demographic attributes.

The majority of facial attribute recognition literature formulates race and gender classification as two separate tasks solved independently. This isolated approach disregards the inherent nature of facial visual identity and fails to explicitly evaluate recognition performance at the intersections of demographic groups, such as Black females or Asian females. Consequently, performance degradations concentrated within specific intersectional subgroups escape the detection of global metrics, thereby limiting a comprehensive understanding of model reliability in real-world deployment. A unified intersectional classification formulation under a balanced evaluation setting provides an objective comparative benchmark to fairly measure performance variations across subgroups. However, intersectional classification modeling demands rich, high-capacity visual representations to distin-

guish decision boundaries between subgroups characterized by a high degree of visual overlap, thereby requiring a visual representation extraction architecture that is robust against phenotypic variations and facial expressions.

The development of facial attribute recognition methodologies has substantially transformed from handcrafted feature extraction approaches toward deep learning and ViT architectures. In the era of Convolutional Neural Network (CNN) architectures, numerous studies explored demographic feature extraction by utilizing specific facial regions such as the central face area [1], multi-scale multi-task frameworks for simultaneous age and gender estimation [2], and deep CNN models for ethnicity recognition [3]. Recent developments demonstrate the superiority of ViTs in modeling global spatial dependencies across image patches via the Multi-Head Self-Attention (MHSA) mechanism, opening avenues for in-depth analysis of visual representation bias [4], while the utilization of synthetic generative representations has also been developed to mitigate demographic disparities [5]. In addition, the integration of parallel attention mechanisms [6], the application of ViTs for facial gender classification [7], and the exploration of multi-axis transformer architectures [8] have further enriched the diversity of visual representation modeling. Furthermore, explorations into combining cross-domain ViT representations have begun through the integration of dual facial and emotional features to enhance the discrimination of race and gender attributes [9], [10]. Nonetheless, the exploration of comprehensive multi-domain representation fusion that simultaneously integrates structural biometric cues, emotional expression dynamics, and facial biological aging morphology still requires an integrated empirical investigation.

Although the exploration of ViTs and vision models has progressed rapidly, three major research gaps remain insufficiently resolved. First, the majority of prior studies rely on single-domain facial biometric representations without simultaneously integrating dynamic affective cues, causing the developed models to struggle in distinguishing subgroups with a high degree of phenotypic overlap. Second, demographic recognition frequently exhibits high sensitivity to facial morphological changes associated with advancing age because the employed feature extraction does not incorporate biological aging representations. Third, comparative evaluations regarding decision boundary behaviors among downstream classifier models encompassing linear, probabilistic, ensemble, and kernel-based models, along with their interactions with feature scaling and dimensionality reduction within the transformer representation space, remain substantially limited. To bridge the limitations of single-domain representations and evaluate these downstream classifier decision boundaries, this study proposes an integration of three ViT representation domains combined with a structured classical classifier pipeline optimization.

To address these research gaps, this study proposes a Tri-Domain ViT feature fusion framework that combines three task-specific latent representations for intersectional

demographic classification from facial images. The proposed framework integrates three complementary representation domains, encompassing facial biometric geometric representations, dynamic affective expression cues, and biological aging morphological characteristics from task-specific pre-trained ViT backbone models into a unified representation via a feature concatenation operation. Latent feature extraction is conducted offline from frozen pre-trained backbone models, thereby preserving computational efficiency while isolating visual representations from the variability of retraining procedures. The resulting feature vectors are subsequently evaluated systematically through a series of single-domain, dual-domain, and tri-domain ablation schemes across four groups of classical machine learning algorithms: RF, GNB, LR, and SVM. All downstream classification architectures are optimized through a Stratified Cross-Validation procedure combined with an isolated modular pipeline of feature scaling and Principal Component Analysis (PCA) dimensionality reduction, ensuring the prevention of data leakage and the establishment of robust decision boundaries across demographic subgroups.

The main scientific contributions of this study are summarized into four primary points as follows:

- 1) A Tri-Domain ViT feature fusion framework integrating face-associated biometric representations, expression-related representations, and age-associated facial representations into a unified latent feature vector for intersectional demographic classification.
- 2) An empirical comparative benchmark across multiple feature ablation schemes and classical machine learning classifiers optimized via Stratified Cross-Validation hyperparameter tuning, examining decision boundary behavior across linear, probabilistic, ensemble, and kernel-based models.
- 3) Competitive classification performance on the Demog-Pairs dataset, achieving higher reported performance compared to the evaluated single-domain and dual-domain configurations on the majority of classifiers.
- 4) A granular intersectional subgroup performance analysis, evaluating subgroup-level classification performance across six demographic subgroups, providing an in-depth characterization of model behaviors across racial and gender intersections while systematically revealing the consistency of multi-domain feature representations on independent evaluation cohorts without exhibiting acute performance degradation.

The organization of this article is structured as follows: Section II reviews related literature on facial demographic recognition and transformer architectures. Section III describes the DemogPairs dataset specifications, multi-domain ViT representation extraction, and the optimized classification pipeline. Section IV presents the experimental evaluation results, feature ablation studies, and intersectional subgroup performance analysis. Finally, Section V summarizes the primary conclusions of the study, methodological limitations,

and recommendations for future research directions.

II. RELATED WORKS

The evolution of facial demographic attribute recognition methodologies has transformed from conventional facial part-based descriptors and handcrafted feature extraction toward deep representation architectures. In the early phase, local region-based approaches exploited middle-face geometry to recognize ethnicity and minimize the interference of background variations [1]. However, classical CNN-based extractors often exhibit sharp accuracy degradation when confronted with extreme illumination variations as well as cross-racial recognition disparities or the Other-Race Effect phenomenon. Increasing the depth of convolutional networks has been proven to enhance visual feature discrimination across diverse populations [3]. In addition, the integration of parallel feature-sharing mechanisms with visual attention was designed to strengthen correlations between facial attributes while simultaneously reducing local representation bias [6].

The application of the ViT architecture in the facial biometrics domain provides visual modeling capabilities that are fundamentally different from conventional approaches. Through the MHSA mechanism, transformer models are capable of capturing global spatial dependencies across facial image patches directly without being constrained by rigid local receptive fields [4]. Such structural characteristics have proven effective in mapping demographic attributes such as gender on standardized-resolution facial images [7]. Alongside standard transformer models, the exploration of hybrid architectures such as MaxViT combines window-based attention and grid attention to process spatial representations hierarchically [8]. Although transformer architectures offer high global representation flexibility, the empirical effectiveness of demographic class separation heavily depends on the quality and characteristics of the training data used.

Investigations into performance disparities in computer vision systems have driven the development of evaluation protocols based on balanced class distributions. Testing on balanced datasets provides controlled experimental conditions to objectively measure accuracy differences across demographic subgroups without distortions from sample size imbalances, although a balanced distribution does not necessarily eliminate social biases inherent in the data. Latent representation analyses demonstrate that transformer models remain susceptible to reproducing social biases originating from large-scale pre-training corpora [4]. To mitigate performance disparities across racial and gender groups, generative augmentation techniques based on StyleGAN2 have been utilized to synthesize viewpoint variations that narrow classification error gaps [5]. Such approaches underscore the importance of comprehensive evaluation across demographic subgroups.

The limitations of single-domain representations in distinguishing facial visual variations have motivated the exploration of multi-scale and multi-task fusion architectures. Multi-task learning approaches have proven capable of ex-

tracting dependencies between age and gender estimation through the simultaneous processing of multiple facial image patches [2]. Parallel representation-sharing mechanisms have also been developed to combine various affective attributes simultaneously [6], while the utilization of generative viewpoints enriches feature variations across demographic groups [5]. Furthermore, the integration of ViT-based dual latent features combining facial biometrics and emotional expressions is capable of enhancing intersectional classification accuracy [9]. Combining biometric features with age representations also demonstrates complementary informational contributions [10], thereby substantiating the empirical potential of fusing cross-domain latent representations.

The selection of downstream classifier models exerts a substantial influence on the effectiveness of decision boundary separation in high-dimensional feature spaces. End-to-end SoftMax classification layers are commonly applied in convolutional neural network architectures to map class probabilities directly [1], [8]. On the other hand, kernel-based algorithms such as SVM and adaptive kernel models exhibit high robustness in separating complex non-linear decision boundaries without requiring exhaustive weight updating processes [3]. In addition, tree-based ensemble models such as RF have proven effective in modeling hierarchical interactions among demographic data through multi-level partitioning [2], whereas probabilistic approaches such as GNB offer high inference efficiency. Such class separation behaviors confirm that each classification model possesses unique decision boundary characteristics.

Based on the reviewed literature, this study occupies a strategic position in comprehensively bridging the gaps in facial demographic attribute modeling. In contrast to prior studies that rely on single-domain representations or partial dual-domain fusions, this study proposes a tri-domain fusion framework that integrates facial biometric geometric representations, dynamic expression affective cues, and biological aging morphological characteristics based on frozen pre-trained ViT models in a unified manner. These multi-domain latent representations are subsequently integrated with an optimized SVM pipeline to form robust non-linear decision boundaries. A comprehensive comparative evaluation is conducted on the balanced DemogPairs dataset to analyze intersectional race and gender classification performance along with its disparity variations.

III. MATERIALS AND METHODS

The proposed framework integrates multi-domain visual representations and machine learning model optimization for intersectional race and gender classification, as illustrated in Figure 1. The methodology begins with partitioning 224×224 pixel resolution facial images from the DemogPairs dataset into training and test partitions via a stratified split. Three frozen pre-trained ViT-Base models extract 768-dimensional latent representations from the [CLS] token offline. The three vectors are combined through concatenation fusion $\mathbf{z}_{\text{tri}} = \mathbf{f}_{\text{face}} \oplus \mathbf{f}_{\text{emotion}} \oplus \mathbf{f}_{\text{age}}$, where $\mathbf{z}_{\text{tri}}$ denotes the 2,304-dimensional

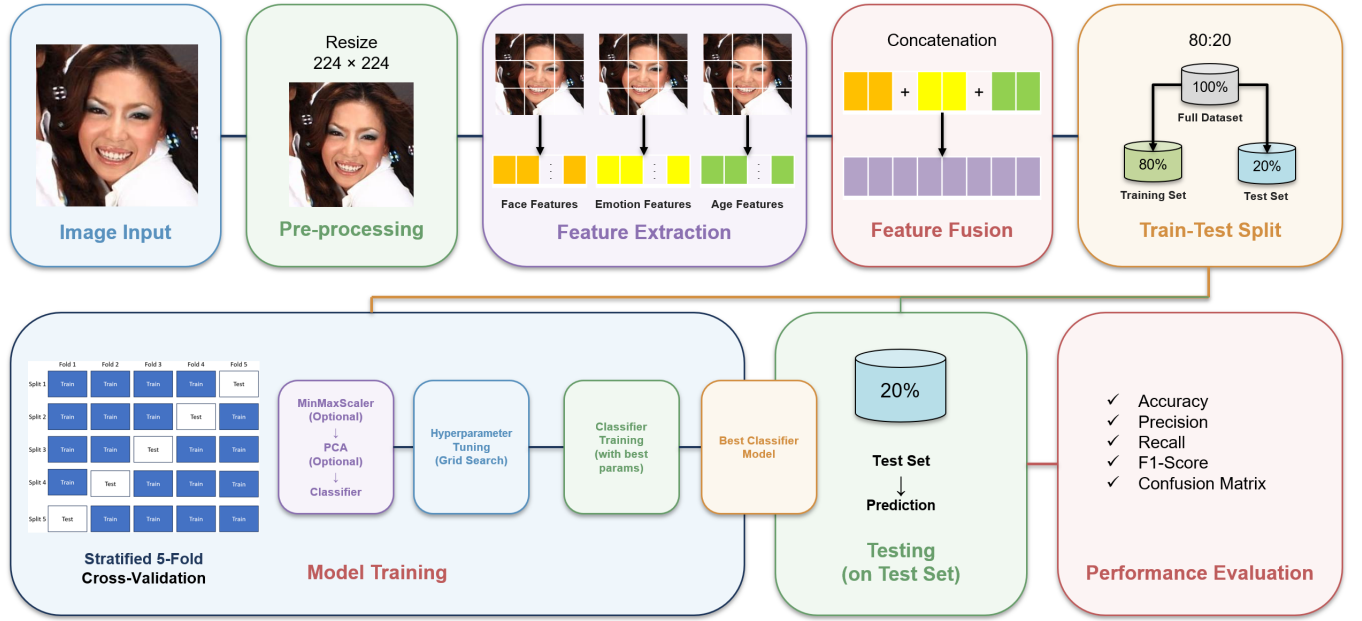


FIGURE 1. End-to-End Framework of the Proposed Multi-Domain Vision Transformer Feature Fusion and Classical Classifier Optimization for Intersectional Race and Gender Classification.

tri-domain vector, $\oplus$ is the concatenation operator, and $\mathbf{f}_{\text{face}}$, $\mathbf{f}_{\text{emotion}}$, and $\mathbf{f}_{\text{age}}$ represent facial biometric, expression, and age features, respectively. The representation vectors are processed within a modular Scaler and PCA pipeline before being classified into six intersectional classes by RF, GNB, LR, and SVM. Hyperparameter tuning is conducted via 5-Fold Stratified Cross-Validation with an isolated GridSearchCV scheme to prevent information leakage prior to test data evaluation.

A. DATASET

The dataset used in this study is DemogPairs [11], a dataset specifically designed to measure facial recognition performance disparities across demographic groups. This dataset contains a total of 10,800 facial images distributed evenly across six intersectional classes. The six subgroups encompass cross-combinations of three macro-race groups and two gender groups, namely Asian Females, Asian Males, Black Females, Black Males, White Females, and White Males, with an exact quota of 1,800 images per class. This balanced distribution provides a controlled evaluation condition to objectively compare performance across subgroups without distortion from majority sample dominance. Visual samples from the six demographic subgroups are illustrated in Figure 2.

To ensure the integrity of empirical testing, the dataset is partitioned using an 80/20 stratified split procedure (80% training data and 20% test data). This partition yields 8,640 training images with 1,440 samples per class and 2,160 independent test images with 360 samples per class, as detailed in Table 1. The test subset is strictly isolated, never involved during the hyperparameter search process, and solely used

TABLE 1. Dataset Partition and Demographic Subgroup Distribution.

Subgroup	Train Set (80%)	Test Set (20%)	Total
Black Males	1,440	360	1,800
White Females	1,440	360	1,800
Asian Males	1,440	360	1,800
White Males	1,440	360	1,800
Black Females	1,440	360	1,800
Asian Females	1,440	360	1,800
Total	8,640	2,160	10,800

for final evaluation. At the image standardization stage, each facial image is converted into a 3-channel Red, Green, Blue (RGB) format, resized to a resolution of 224×224 pixels, and its pixel intensity is scaled from the range $[0, 255]$ to $[0, 1]$ to align with the transformer model inputs.

B. VISION TRANSFORMER

The visual representation extraction architecture of this study adopts ViT-Base (vit-base-patch16-224), as illustrated in Figure 3. The input facial image of 224×224 pixels with three color channels is partitioned into 196 non-overlapping spatial patches $\mathbf{x}_p^i$ of size 16×16 pixels. Each patch is linearly projected via matrix $\mathbf{E}$ into a latent space of dimension $D = 768$, and then concatenated with a class token $\mathbf{x}_{\text{class}}$ and positional embedding $\mathbf{E}_{\text{pos}}$ according to (1), where $\mathbf{z}_0$ represents the initial token sequence vector input to the encoder:

$$\mathbf{z}_0 = [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \dots; \mathbf{x}_p^{196} \mathbf{E}] + \mathbf{E}_{\text{pos}} \quad (1)$$

The initial token sequence vector $\mathbf{z}_0$ is passed to a stack of 12 identical transformer encoder layers. Each encoder layer ℓ processes the input $\mathbf{z}_{\ell-1}$ into an intermediate representation



FIGURE 2. Sample Images of the DemogPairs Dataset across Six Intersectional Demographic Subgroups: (a) Asian Females, (b) Asian Males, (c) Black Females, (d) Black Males, (e) White Females, and (f) White Males.

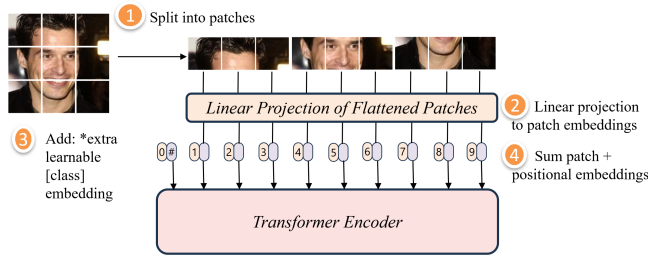


FIGURE 3. Architecture of the ViT Backbone and Patch Projection.

$\mathbf{z}'_\ell$ via the Multi-Head Self-Attention (MHSA) mechanism following the Layer Normalization (LN) operation according to (2), where $\text{LN}(\cdot)$ denotes layer normalization and $\mathbf{z}_{\ell-1}$ is the output representation from the previous layer:

$$\mathbf{z}'_\ell = \text{MHSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1} \quad (2)$$

This process continues toward the output representation $\mathbf{z}_\ell$ via a two-layer Multi-Layer Perceptron (MLP) block with GeLU activation according to (3), where $\text{MLP}(\cdot)$ denotes the multi-layer perceptron network and $\mathbf{z}'_\ell$ is the residual intermediate representation:

$$\mathbf{z}_\ell = \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell \quad (3)$$

The self-attention mechanism with 12 attention heads directly maps global spatial relationships among facial patches without local receptive field constraints.

To maintain computational efficiency and prevent re-training variability, the three ViT backbone models are frozen as offline providers of task-associated representations. The ViT-Face model (skutaada/VITVGGFace) captures representations related to facial biometric geometry, ViT-Emotion (dima806/facial_emotions_image_detection) yields representations related to facial expressions, and ViT-Age (dima806/facial_age_image_detection) extracts representations related to facial age estimation. From each domain model, the feature vector $\mathbf{f}_{\text{domain}} \in \mathbb{R}^{768}$ is extracted from the [CLS] token representation at the final encoder layer L ($\mathbf{z}_L^0$) following the LN operation according to (4), where $\mathbf{z}_L^0$ denotes the first token representation vector at layer L :

$$\mathbf{f}_{\text{domain}} = \text{LN}(\mathbf{z}_L^0) \in \mathbb{R}^{768} \quad (4)$$

The three single-domain latent representations are combined into a tri-domain fusion vector $\mathbf{z}_{\text{tri}} \in \mathbb{R}^{2304}$ via the feature concatenation operation ($\oplus$) of vectors $\mathbf{f}_{\text{face}}$, $\mathbf{f}_{\text{emotion}}$, and $\mathbf{f}_{\text{age}}$, as formulated in (5), where $\mathbf{z}_{\text{tri}}$ denotes the tri-domain fused representation vector:

$$\mathbf{z}_{\text{tri}} = \mathbf{f}_{\text{face}} \oplus \mathbf{f}_{\text{emotion}} \oplus \mathbf{f}_{\text{age}} \in \mathbb{R}^{2304} \quad (5)$$

TABLE 2. Multi-Domain Feature Fusion and Ablation Configurations.

#	Configuration	Domain Category	Dimension
1	Face	Single-Domain	768
2	Emotion	Single-Domain	768
3	Age	Single-Domain	768
4	Emotion $\oplus$ Face	Dual-Domain	1,536
5	Face $\oplus$ Age	Dual-Domain	1,536
6	Emotion $\oplus$ Age	Dual-Domain	1,536
7	Face $\oplus$ Emotion $\oplus$ Age	Tri-Domain (Proposed)	2,304

The systematic exploration encompasses seven feature ablation schemes, comprising three single-domain configurations of dimension 768, three dual-domain configurations of dimension 1,536, and one tri-domain configuration of dimension 2,304, as summarized in Table 2.

C. RANDOM FOREST

The RF ensemble model aggregates predictions from a collection of independent decision trees to classify facial features based on the principles of bootstrap aggregating (bagging) and the random subspace method. This approach reduces prediction variance by constructing B classification trees from random samples of the training data. Each node split restricts feature selection to a random subset of size $\text{max_features} \in \{\text{'sqrt'}, \text{'log2'}\}$ to suppress inter-tree correlation. The split criterion optimizes the Gini impurity index until satisfying the minimum node split threshold min_samples_split , the maximum depth limit max_depth , or the minimum leaf threshold min_samples_leaf . The majority voting mechanism determines the final decision for an image sample across all decision trees. The hyperparameter tuning procedure evaluates combinations of feature scaling, PCA dimensionality reduction, variations in the number of trees n_estimators , and tree growth constraints, yielding 288 grid configurations

TABLE 3. Hyperparameter Search Space for Random Forest Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Number of Estimators	100, 200	2
Maximum Tree Depth	None, 20, 30	3
Feature Subspace	'sqrt', 'log2'	2
Minimum Samples Split	2, 5	2
Minimum Samples Leaf	1, 2	2
Total Grid Combinations	–	288 (1,440 fits)

with 1,440 total fits via 5-Fold Stratified Cross-Validation as summarized in Table 3.

D. GAUSSIAN NAIVE BAYES

Based on Bayes' theorem, the GNB classifier estimates the posterior probability of intersectional demographic classes under the assumption that all continuous feature dimensions are conditionally independent. The probability density function of the normal distribution models the likelihood of continuous features with respect to the target class according to the formulation in (6), where x_i denotes the value of the i -th continuous feature, y is the class label variable, c represents the target class category, and $\mu_{c,i}$ and $\sigma_{c,i}^2$ represent the mean and variance of the i -th feature in class c , respectively:

$$P(x_i | y = c) = \frac{1}{\sqrt{2\pi\sigma_{c,i}^2}} \exp\left(-\frac{(x_i - \mu_{c,i})^2}{2\sigma_{c,i}^2}\right) \quad (6)$$

The addition of the smoothing parameter `var_smoothing` maintains computational stability against the risk of near-zero variance in high-dimensional latent representation spaces. The hyperparameter tuning procedure via 5-Fold Stratified Cross-Validation evaluates combinations of data scaling, PCA dimensionality reduction, and 40 `var_smoothing` intervals explored logarithmically from 1.0×10^{-9} to 1.0×10^2 , yielding 240 grid combinations with 1,200 total fits as presented in Table 4.

E. LOGISTIC REGRESSION

The multinomial regression approach (Softmax regression) in LR maps latent feature representations into six intersectional demographic categories through unified linear probability estimation. The Softmax probability function calculates the conditional probability of a sample with respect to the target category as defined in (7), where $\mathbf{x}$ represents the input feature vector, y is the class label variable, c denotes the target class category, $\mathbf{w}_c$ and b_c represent the weight vector and bias of class c , respectively, while the constant $K = 6$ indicates the total number of intersectional classes:

$$P(y = c | \mathbf{x}) = \frac{e^{\mathbf{w}_c^T \mathbf{x} + b_c}}{\sum_{j=1}^K e^{\mathbf{w}_j^T \mathbf{x} + b_j}} \quad (7)$$

The model training process minimizes the cross-entropy loss function by incorporating an L_2 regularization penalty

to control weight complexity in the high-dimensional representation space. The structured hyperparameter search evaluates variations in the regularization strength C , optimization solver algorithms (lbfgs, saga, newton-cg), iteration limit `max_iter`, as well as the integration of scaling and PCA, yielding 270 evaluation combinations or 1,350 total fits under the 5-Fold Stratified Cross-Validation protocol presented in Table 5.

F. SUPPORT VECTOR MACHINE

The SVM classifier constructs a maximum margin hyperplane to separate the six intersectional classes in the high-dimensional latent representation space. Kernel functions project the nonlinear data distribution from multi-domain fusion into a higher-dimensional feature space, encompassing linear, Radial Basis Function (RBF), and polynomial kernel options. The second-degree polynomial kernel formulation in (8) computes the dot product $\langle \mathbf{x}_i, \mathbf{x}_j \rangle$ between pairs of feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$, where γ denotes the kernel scaling parameter, $d = 2$ determines the mapping degree, and the intercept constant `coef0` = 0.0 follows the scikit-learn default value:

$$K(\mathbf{x}_i, \mathbf{x}_j) = (\gamma \langle \mathbf{x}_i, \mathbf{x}_j \rangle + \text{coef0})^d, \quad d = 2 \quad (8)$$

This quadratic mapping facilitates the efficient capture of nonlinear interactions across visual dimensions.

Hyperparameter optimization evaluates penalty parameter ranges $C \in \{0.01, 0.1, 1, 10\}$, various kernel functions, γ coefficients, polynomial degrees, as well as scaling and PCA options, yielding 288 configuration combinations or 1,440 total fits across the 5-Fold Stratified Cross-Validation procedure as presented in Table 6.

G. CLASSIFICATION PIPELINE

The modular pipeline framework unifies feature scaling, representational dimensionality reduction, and downstream model estimation into an integrated transformation chain. This processing chain is formulated in (9), where the input feature vector $\mathbf{x}$ is transformed by the scaling module into $\tilde{\mathbf{x}}$, dimensionally reduced via PCA into $\hat{\mathbf{x}}$, and mapped by the classifier to the predicted label $\hat{y} \in \{0, 1, 2, 3, 4, 5\}$ for the six demographic subgroups:

$$\mathbf{x} \xrightarrow{\text{Scaler}} \tilde{\mathbf{x}} \xrightarrow{\text{PCA}} \hat{\mathbf{x}} \xrightarrow{\text{Classifier}} \hat{y} \in \{0, 1, 2, 3, 4, 5\} \quad (9)$$

TABLE 4. Hyperparameter Search Space for Gaussian Naive Bayes Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Variance Smoothing	logspace($-9, 2, 40$) (1.0×10^{-9} to 1.0×10^2)	40
Total Grid Combinations	–	240 (1,200 fits)

TABLE 5. Hyperparameter Search Space for Logistic Regression Classifier.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Regularization Strength	0.01, 0.1, 1, 10, 100	5
Optimization Solver	'lbfgs', 'saga', 'newton-cg'	3
Maximum Iterations	500, 1000, 2000	3
Total Grid Combinations	–	270 (1,350 fits)

TABLE 6. Hyperparameter Search Space for Support Vector Machine.

Component / Hyperparameter	Evaluated Values	Count
Feature Scaler	None, MinMaxScaler	2
Dimensionality Reduction (PCA)	None, 0.50, 0.75	3
Regularization Parameter	0.01, 0.1, 1, 10	4
Kernel Function	'linear', 'rbf', 'poly'	3
Polynomial Degree	2, 3	2
Kernel Coefficient	'scale', 'auto'	2
Total Grid Combinations	–	288 (1,440 fits)

The cross-validation protocol implements a 5-Fold Stratified Cross-Validation scheme with controlled shuffling to maintain a balanced proportion of $\sim 16.67\%$ per class in each fold. To prevent information leakage, the Scaler and PCA are fitted exclusively on the training data within GridSearchCV before being applied to the validation data. The search over 1,086 hyperparameter combinations yields 5,430 fits per feature scheme or 38,010 total fits for the seven representation configurations, along with 28 final refits prior to testing on the independent held-out test data.

H. EVALUATION METRICS

The performance evaluation of the multi-class classification for the six demographic subgroups at the subgroup level is conducted using a binary One-vs-Rest (OvR) scheme. This scheme utilizes four per-class confusion matrix components to compute OvR Accuracy, Precision, Recall, and F1-Score for each class $c \in \{1, \dots, K\}$. These four components comprise TP_c (True Positive), TN_c (True Negative), FP_c (False Positive), and FN_c (False Negative). OvR Accuracy in (10) measures the ratio of total correct predictions to the entire test data:

$$\text{Accuracy}_c = \frac{TP_c + TN_c}{TP_c + TN_c + FP_c + FN_c} \quad (10)$$

Meanwhile, Precision in (11) and Recall in (12) map the positive accuracy and detection sensitivity of the system per

subgroup:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (11)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (12)$$

The F1-Score in (13) is formulated as the harmonic mean of Precision and Recall to represent the balanced evaluation of detection performance per subgroup:

$$\text{F1-Score}_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (13)$$

To evaluate the overall effectiveness of the classification system on independent test data, subgroup-level performance metrics are aggregated into global evaluation metrics. In this formulation, the symbol N represents the total test data samples ($N = 2,160$), while K denotes the total number of demographic subgroups ($K = 6$). Global Accuracy (Overall Accuracy) in (14) is calculated as the ratio of total correct predictions, namely the sum of the main diagonal elements of the confusion matrix $\sum_{c=1}^K TP_c$, to the total test samples N :

$$\text{Accuracy}_{\text{Global}} = \frac{\sum_{c=1}^K TP_c}{N} \quad (14)$$

Furthermore, global Precision (Macro Precision) in (15), global Recall (Macro Recall) in (16), and global F1-Score (Macro F1-Score) in (17) are computed via an unweighted

macro average across the $K = 6$ subgroups to assign equal evaluation weight to each demographic class:

$$\text{Precision}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{Precision}_c \quad (15)$$

$$\text{Recall}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{Recall}_c \quad (16)$$

$$\text{F1-Score}_{\text{Global}} = \frac{1}{K} \sum_{c=1}^K \text{F1-Score}_c \quad (17)$$

IV. RESULTS AND DISCUSSION

A. GLOBAL PERFORMANCE ANALYSIS

The performance evaluation of RF across the seven feature configurations on the DemogPairs test data is presented in Table 7. Empirical experimental results demonstrate that the dual-domain Emotion $\oplus$ Face scheme achieves the highest performance, with an Accuracy of 86.85% and an F1-Score of 86.82%. Conversely, the tri-domain concatenation Face $\oplus$ Emotion $\oplus$ Age exhibits a slight decrease in performance to an Accuracy of 86.20% and an F1-Score of 86.13%. This decline is interpreted as a potential challenge in partitioning the 2,304-dimensional feature space through random tree splits (random splits). All RF configurations consistently select PCA 0.75 dimensionality reduction to balance data variance. In the single-domain category, the Face feature records the best performance with an Accuracy of 85.46%, surpassing the Emotion domain feature at 80.60% as well as the Age domain at 73.66%.

The performance evaluation of the probabilistic GNB classification across seven feature configuration variations is summarized in Table 8. Although constrained by the fundamental assumption of continuous feature independence, the GNB model demonstrates a consistent performance improvement trend from the single-domain to multi-domain categories. In the single-domain category, the Age configuration produces the lowest performance with an Accuracy of 69.63% and an F1-Score of 69.52%, whereas the Face feature achieves an Accuracy of 82.69%. Feature integration in the dual-domain category consistently enhances accuracy, led by the Emotion $\oplus$ Face combination at 84.86%. The peak GNB classification performance is attained by the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age with an Accuracy of 85.05% and an F1-Score of 85.05%. All GNB feature configurations uniformly select PCA 0.75 dimensionality reduction to minimize correlations among latent features.

The benchmark results of multinomial LR performance across all feature configurations are presented in Table 9. A key characteristic of the LR model is the retention of the full feature representation, in which all schemes (seven out of seven configurations) select no PCA reduction (pca=None). Empirical evaluation indicates that the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age records the highest performance with an Accuracy of 92.73% and an F1-Score of 92.73% via the penalty parameter $C = 0.1$ and the newton-cg solver.

This tri-domain achievement surpasses the best dual-domain configuration Emotion $\oplus$ Face, which attains an Accuracy of 92.41% and an F1-Score of 92.40%, as well as outperforms the best single-domain configuration Face with an Accuracy of 90.60%. These findings indicate the effectiveness of L_2 regularization in handling high-dimensional feature spaces without principal component compression.

The performance progression of SVM across all feature ablation stages is summarized in Table 10. Similar to the linear model, all SVM configurations consistently maintain the full representation space without PCA reduction (pca=None). Empirical evaluation reveals a tiered performance increase, commencing from the best single-domain configuration Face with an Accuracy of 90.83% and an F1-Score of 90.83%. Representation integration in the dual-domain scheme Emotion $\oplus$ Face enhances performance to an Accuracy of 93.29% and an F1-Score of 93.29%, before ultimately reaching the highest point in the tri-domain configuration Face $\oplus$ Emotion $\oplus$ Age with an Accuracy of 93.70% and an F1-Score of 93.69%. In hyperparameter tuning, the degree parameter with a value of 2 is only active for the poly kernel function, whereas RBF kernel-based models do not activate this parameter.

A descriptive comparative synthesis across the 28 experimental combinations shows that the SVM model based on the tri-domain fusion Face $\oplus$ Emotion $\oplus$ Age occupies the highest performance achievement within the evaluated search space, attaining an Accuracy of 93.70% and an F1-Score of 93.69%. Based on the average classifier performance across all feature configurations, the descriptive hierarchical ranking of algorithmic effectiveness is led by SVM (91.47%), followed by LR (90.40%), RF (82.81%), and GNB (79.37%). The non-linear characteristics of the polynomial kernel in SVM prove effective in mapping decision boundaries between high-dimensional demographic subgroups. However, this performance ranking comparison is strictly empirical and descriptive within the controlled experimental setup, and does not reflect absolute statistical superiority among algorithm families without formal hypothesis testing.

B. FEATURE ABLATION ANALYSIS

The quantification of classification accuracy changes from single-domain schemes toward dual- and tri-domain configurations is summarized in Table 7, Table 8, Table 9, and Table 10. In the SVM model, the representation transition from Face (90.83%) to Emotion $\oplus$ Face (93.29%) increases accuracy by 2.46% (0.0246). The addition of a third domain in the tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration yields a further performance improvement to 93.70%, which reflects a cumulative increase of 2.87% (0.0287) over the single Face feature. This pattern of progressive improvement indicates the presence of additional discriminative information from each combined visual representation domain, although the interaction mechanisms among latent feature dimensions cannot be definitively confirmed solely from empirical test results.

An analysis of the contribution of the age domain (Age) representation dissects the role of latent aging information

TABLE 7. Performance Benchmark of Random Forest across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	0.8546	0.8543	0.8546	0.8539	n_est=200, depth=30, max_feat=log2, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=MinMaxScaler
Emotion	0.8060	0.8063	0.8060	0.8057	n_est=200, depth=None, max_feat=log2, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None
Age	0.7366	0.7363	0.7366	0.7354	n_est=200, depth=30, max_feat=log2, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=None
Emotion $\oplus$ Face	0.8685	0.8689	0.8685	0.8682	n_est=200, depth=None, max_feat=sqrt, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None
Face $\oplus$ Age	0.8579	0.8578	0.8579	0.8573	n_est=200, depth=None, max_feat=sqrt, min_split=2, min_leaf=1, pca=PCA(0.75), scaler=None
Emotion $\oplus$ Age	0.8111	0.8111	0.8111	0.8108	n_est=200, depth=None, max_feat=log2, min_split=5, min_leaf=2, pca=PCA(0.75), scaler=None
Face $\oplus$ Emotion $\oplus$ Age	0.8620	0.8620	0.8620	0.8613	n_est=200, depth=30, max_feat=sqrt, min_split=5, min_leaf=1, pca=PCA(0.75), scaler=None

TABLE 8. Performance Benchmark of Gaussian Naive Bayes across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	0.8269	0.8271	0.8269	0.8258	var_smoothing=4.1246e-02, pca=PCA(0.75), scaler=MinMaxScaler
Emotion	0.7338	0.7387	0.7338	0.7329	var_smoothing=3.0703e-03, pca=PCA(0.75), scaler=None
Age	0.6963	0.6979	0.6963	0.6952	var_smoothing=4.3755e-04, pca=PCA(0.75), scaler=MinMaxScaler
Emotion $\oplus$ Face	0.8486	0.8490	0.8486	0.8481	var_smoothing=5.8780e-03, pca=PCA(0.75), scaler=MinMaxScaler
Face $\oplus$ Age	0.8315	0.8343	0.8315	0.8317	var_smoothing=1.1253e-02, pca=PCA(0.75), scaler=MinMaxScaler
Emotion $\oplus$ Age	0.7681	0.7686	0.7681	0.7681	var_smoothing=1.6037e-03, pca=PCA(0.75), scaler=MinMaxScaler
Face $\oplus$ Emotion $\oplus$ Age	0.8505	0.8512	0.8505	0.8505	var_smoothing=5.8780e-03, pca=PCA(0.75), scaler=None

within the demographic classification system. Across all evaluated classifiers, Age constitutes the single-domain configuration with the lowest accuracy attainment, as recorded at 87.64% for SVM and 69.63% for GNB. Nevertheless, combining the Age feature with facial biometric features in the dual-domain Face $\oplus$ Age scheme yields an accuracy of 92.55%, providing an improvement of 1.72% (0.0172) over the pure Face feature performance (90.83%) on the SVM model. These empirical results indicate the likely presence of additional discriminative information from age-associated representations that complements the facial morphological profile, without claiming that such informational contribution is orthogonal or strictly complementary theoretically.

C. INTERSECTIONAL SUBGROUP PERFORMANCE ANALYSIS

The granular classification performance profile of the best-performing SVM model based on the tri-domain Face $\oplus$ Emotion $\oplus$ Age fusion is thoroughly evaluated based on OvR metrics as presented in Table 11. All demographic subgroups successfully achieve an F1-Score exceeding 91.00%, with performance ranges spanning from 91.74% for Black Females to 96.14% for White Males, resulting in a range disparity difference of 4.40% (0.0440). Meanwhile, OvR Accuracy values fall within the range of 97.31% to 98.70% with

a disparity of 1.39 pp. It should be noted methodologically that the high OvR Accuracy values are also influenced by the 5:1 negative sample dominance ratio in the binary OvR evaluation, such that this metric is not positioned as the sole indicator of performance disparity.

A comparison of subgroup performance profiles between the SVM model and the linear LR classifier under the tri-domain configuration is presented in Table 11. The tri-domain LR model likewise demonstrates high performance consistency, with F1-Score achievements exceeding 91.00% across all demographic classes, spanning from 91.36% for Asian Females to 95.58% for White Males with a range disparity difference of 4.22% (0.0422). The White Males subgroup consistently records the highest classification performance across both classifiers. These empirical findings demonstrate that the tri-domain representation fusion yields robust visual representations that are stably distributed across all demographic subgroups, in both linear and non-linear models. Nevertheless, this subgroup evaluation reflects granular per-class performance rather than a comprehensive algorithmic fairness audit.

D. ERROR PATTERN ANALYSIS

The shift in classification error distributions from the single-domain scheme toward dual- and tri-domain configurations is

TABLE 9. Performance Benchmark of Logistic Regression across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	0.9060	0.9060	0.9060	0.9059	$C = 1$, solver=newton-cg, max_iter=500, pca=None, scaler=MinMaxScaler
Emotion	0.8847	0.8850	0.8847	0.8846	$C = 1$, solver=saga, max_iter=500, pca=None, scaler=MinMaxScaler
Age	0.8648	0.8649	0.8648	0.8648	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Emotion $\oplus$ Face	0.9241	0.9241	0.9241	0.9240	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Face $\oplus$ Age	0.9162	0.9162	0.9162	0.9162	$C = 0.1$, solver=newton-cg, max_iter=500, pca=None, scaler=None
Emotion $\oplus$ Age	0.9051	0.9052	0.9051	0.9051	$C = 0.1$, solver=lbfgs, max_iter=500, pca=None, scaler=None
Face $\oplus$ Emotion $\oplus$ Age	0.9273	0.9275	0.9273	0.9273	$C = 0.1$, solver=newton-cg, max_iter=500, pca=None, scaler=None

TABLE 10. Performance Benchmark of Support Vector Machine across Seven Feature Configurations.

Configuration	Accuracy	Precision	Recall	F1-Score	Best Parameters
Face	0.9083	0.9084	0.9083	0.9083	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Emotion	0.9019	0.9020	0.9019	0.9017	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Age	0.8764	0.8767	0.8764	0.8765	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Emotion $\oplus$ Face	0.9329	0.9333	0.9329	0.9329	$C = 10$, rbf, γ =scale, pca=None, scaler=MinMaxScaler
Face $\oplus$ Age	0.9255	0.9254	0.9255	0.9254	$C = 10$, poly, γ =scale, deg=2, pca=None, scaler=None
Emotion $\oplus$ Age	0.9208	0.9210	0.9208	0.9209	$C = 10$, rbf, γ =scale, pca=None, scaler=None
Face $\oplus$ Emotion $\oplus$ Age	0.9370	0.9372	0.9370	0.9369	$C = 10$, poly, γ =scale, deg=2, pca=None, scaler=None

TABLE 11. Subgroup-Level Performance for Tri-Domain Models.

Classifier	Subgroup	Recall	Precision	F1-Score	OvR Accuracy
SVM (Tri)	White Males	0.9694	0.9536	0.9614	98.70%
	Black Males	0.9417	0.9549	0.9483	98.29%
	White Females	0.9472	0.9241	0.9355	97.82%
	Asian Males	0.9444	0.9239	0.9341	97.78%
	Asian Females	0.9250	0.9250	0.9250	97.50%
	Black Females	0.8944	0.9415	0.9174	97.31%
LR (Tri)	White Males	0.9611	0.9505	0.9558	98.52%
	Black Males	0.9306	0.9571	0.9437	98.15%
	White Females	0.9222	0.9121	0.9171	97.22%
	Asian Males	0.9278	0.9076	0.9176	97.22%
	Asian Females	0.9111	0.9162	0.9136	97.13%
	Black Females	0.9111	0.9213	0.9162	97.22%

analyzed through the confusion matrix diagrams in Figure 4, which encompass Figure 4(a), Figure 4(b), and Figure 4(c). Empirical evaluation reveals that the total misclassifications across the 2,160 held-out test data consistently decrease as the latent representation domains expand. The single-domain Face SVM model yields 198 prediction errors, which subsequently decrease to 145 samples in the dual-domain Emotion $\oplus$ Face scheme. The tri-domain Face $\oplus$ Emotion $\oplus$ Age configuration suppresses classification errors to the lowest point of 136 samples. This consistent reduction in errors

indicates that multi-domain representation fusion sharpens the separation boundaries between demographic subgroups in the latent feature space.

An in-depth examination of the confusion matrix for the single-domain Face model in Figure 4(a) reveals that classification errors are predominantly concentrated in inter-racial subgroups within the same gender. This tendency is particularly pronounced in the female cohort, where the Asian Females subgroup incurs 46 misclassifications, encompassing 16 samples predicted as White Females and 16 samples

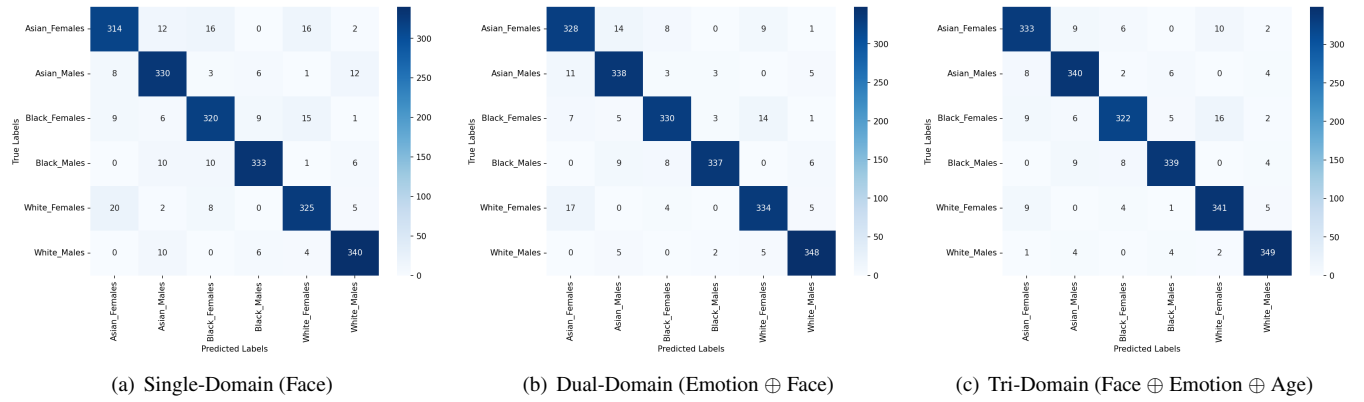


FIGURE 4. Confusion Matrices across Feature Fusion Schemes on the Held-Out Test Set: (a) Single-Domain (Face), (b) Dual-Domain (Emotion ⊕ Face), and (c) Tri-Domain (Face ⊕ Emotion ⊕ Age).

as Black Females. Conversely, the White Females subgroup records 20 prediction errors distributed to Asian Females. This empirical pattern is consistent with the potential presence of inter-racial phenotypic overlap within single facial biometric geometric representations. Nevertheless, such confusion matrix evidence does not prove that visual overlap is the sole cause of all observed misclassifications.

The shift in classification error distributions in the dual-domain Emotion ⊕ Face model is further analyzed based on the confusion matrix in Figure 4(b). The addition of affective expression latent representations effectively reduces the inter-racial misclassification pattern among same-gender subgroups, particularly within the female group. Prediction errors from Asian Females to White Females are substantially suppressed from 16 cases to 9 cases. Concurrently, the number of correctly predicted samples (true positives) in the White Females subgroup increases from 325 images to 334 images. The integration of dynamic facial expression signals strengthens the inter-racial visual discriminant boundaries within the latent representation space, thereby enabling the classifier to distinguish similar morphological features with a more robust and stable level of certainty.

The misclassification patterns in the tri-domain Face ⊕ Emotion ⊕ Age configuration are comprehensively evaluated based on the confusion matrix in Figure 4(c). The integration of age-domain representations consistently increases the number of correct predictions in the Asian Females subgroup to 333 samples and White Females to 341 samples. Furthermore, cross-gender misclassifications, representing instances where female subjects are confused as male or vice versa, amount to only 51 cases out of the 2,160 held-out test data (2.36%). These empirical findings indicate that inter-gender boundaries are separated very distinctly within the multi-domain latent representation space, such that the remaining classification errors in the tri-domain model are almost entirely dominated by inter-racial misclassifications within the same gender group.

E. ANALYSIS OF SELECTED SVM KERNEL CONFIGURATION

Hyperparameter optimization via GridSearchCV established the second-degree polynomial kernel-based SVM model as the best configuration for intersectional race and gender classification. This selected configuration applies a penalty parameter of $C = 10$ without PCA or Scaler, and adopts the polynomial kernel function in (8) with a scaling coefficient of $\gamma = \text{scale}$, a default constant of $\text{coef0} = 0.0$, and a mapping degree of $d = 2$. This empirical finding affirms that the selected configuration is the highest-performing model within the evaluated search space, rather than a claim of absolute mathematical superiority of the kernel function in general. Conceptually, the second-degree polynomial mapping enables the model to capture quadratic interactions among multi-domain latent feature dimensions without requiring an explicit infinite-dimensional projection, thereby making the resulting decision boundary more flexible in separating demographic subgroups.

Another important characteristic of the best SVM configuration in the empirical evaluation is the consistent selection of the full feature representation space without PCA dimensionality reduction ($\text{pca}=\text{None}$). The selection of the $\text{pca}=\text{None}$ option indicates that retaining all 2,304 dimensions of the tri-domain latent representation provides the best classification performance within the evaluated search space. However, this interpretation is entirely grounded in empirical observational evidence, so the test results cannot be concluded as definitive proof that PCA reduction discards crucial discriminative information. The regularization characteristic of SVM through margin maximization proves effective in managing high-dimensional feature spaces without suffering performance degradation due to overfitting, thereby comprehensively preserving the latent geometric structure of the three facial representation domains.

F. COMPARISON WITH PRIOR STUDIES

A direct empirical comparison between the proposed framework and prior studies evaluated on the DemogPairs dataset is summarized in Table 12. The proposed model based on

TABLE 12. Comparative Performance of Proposed Framework against Prior Studies on the DemogPairs Dataset.

Model	Accuracy	Precision	Recall	F1-Score
MD-ViT [10]	89.07%	0.8912	0.8907	0.8901
Dual-ViT [9]	92.41%	0.9248	0.9241	0.9238
Ours (Tri-Domain ViT + SVM)	93.70%	0.9372	0.9370	0.9369

Tri-Domain ViT + SVM (Face $\oplus$ Emotion $\oplus$ Age) achieves an Accuracy of 93.70%, a Precision of 0.9372, a Recall of 0.9370, and an F1-Score of 0.9369. These quantitative achievements surpass the metrics reported for the Dual-ViT + SVM model by Putri *et al.* [9], namely an Accuracy of 92.41% and an F1-Score of 0.9238, as well as the MD-ViT + XGBoost model by Putri *et al.* [10] with an Accuracy of 89.07% and an F1-Score of 0.8901. All comparative figures are directly cited from their respective publications, such that this comparison serves as a contextual benchmark on the same reference dataset and is not intended as an identical experimental replication.

The seminal research on the design of the DemogPairs dataset by Hupont and Fernández [11] established an essential foundation for measuring the impact of demographic imbalance on facial recognition systems. In response to this challenge, the three-domain representation integration framework proposed in this study achieves the highest accuracy among compared studies on the DemogPairs dataset. In addition to evaluating global aggregate metrics, this investigation conducts a comprehensive subgroup disparity analysis on the best tri-domain configuration to transparently map performance variations across demographic subgroups. This detailed evaluation demonstrates recognition stability across intersectional race and gender categories, although a cross-scheme disparity comparison with all prior studies could not be included because such performance variation information was not reported in the related literature.

V. CONCLUSION

This study evaluates a Tri-Domain ViT latent representation fusion framework (Face $\oplus$ Emotion $\oplus$ Age: 2,304 dimensions) for intersectional race and gender classification on the DemogPairs dataset. Experimental results demonstrate that tri-domain fusion achieves the best configuration across 3 of the 4 evaluated classifiers, encompassing SVM, LR, and GNB. Within the evaluated search space, the optimized SVM model with hyperparameter $C = 10$ and a degree-two polynomial kernel yields the highest classification performance, namely an Accuracy of 93.70% and an F1-Score of 93.69% (0.9369). Nevertheless, the benefit of multi-domain representation integration proves to be classifier-dependent, as RF achieves its peak performance on the dual-domain Emotion $\oplus$ Face scheme with an accuracy of 86.85%. In addition, subgroup performance evaluation confirms that the F1-Score values across all six demographic subgroups are stably distributed within the range of 91.74% to 96.14% (0.9174 to 0.9614).

Despite these achievements, this study has several limitations that provide opportunities for future research. First, the demographic scope of this investigation is limited to three macro-racial groups (Asian, Black, and White) in the DemogPairs dataset, so the obtained empirical findings cannot be directly generalized to multiracial populations or other ethnic groups without additional comprehensive testing. Second, latent representation extraction is conducted separately (offline) utilizing frozen pre-trained ViT backbones without simultaneous end-to-end fine-tuning. Third, the experimental evaluation remains focused on controlled laboratory environment images, such that the model's response to extreme pose variations, occlusions, and unconstrained (in-the-wild) illumination requires further investigation. Therefore, future research directions are recommended to explore cross-modal adaptive attention fusion mechanisms (cross-attention transformer fusion) trained end-to-end to learn dynamic cross-domain interaction weights, as well as extending evaluation to larger, more heterogeneous datasets such as FairFace or UTKFace. In addition, the implementation of visual explainability methods such as Attention Rollout or Grad-CAM is crucial to transparently map the spatial contribution of each domain, accompanied by the enforcement of ethical artificial intelligence governance (Responsible AI) with human-in-the-loop oversight to prevent exploitation in public surveillance systems.

REFERENCES

- [1] D. Belcar, P. Grd, and I. Tomićić, "Automatic ethnicity classification from middle part of the face using convolutional neural networks," *Informatics*, vol. 9, no. 1, p. 18, 2022.
- [2] H. Liao, L. Yuan, M. Wu, L. Zhong, G. Jin, and N. Xiong, "Face gender and age classification based on multi-task, multi-instance and multi-scale learning," *Applied Sciences*, vol. 12, no. 23, p. 12432, 2022.
- [3] G. Sunitha, K. Geetha, S. Neelakandan, A. K. S. Pundir, S. Hemalatha, and V. Kumar, "Intelligent deep learning based ethnicity recognition and classification using facial images," *Image and Vision Computing*, vol. 121, p. 104404, 2022.
- [4] J. Brinkmann, P. Swoboda, and C. Bartelt, "A multidimensional analysis of social biases in vision transformers," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 4891–4900.
- [5] S. Ramachandran and A. Rattani, "Deep generative views to mitigate gender classification bias across gender-race groups," in *Pattern Recognition, Computer Vision, and Image Processing. ICPR 2022 International Workshops and Challenges*, J.-J. Rousseau and B. Kapralos, Eds. Cham, Switzerland: Springer Nature Switzerland, 2023, pp. 551–569.
- [6] S. Chen, X. Lai, Y. Yan, D.-H. Wang, and S. Zhu, "Learning an attention-aware parallel sharing network for facial attribute recognition," *Journal of Visual Communication and Image Representation*, vol. 90, p. 103745, 2023.
- [7] G. G. Tahyudin, M. D. Sulistiyo, M. Arzaki, and E. Rachmawati, "Classifying gender based on face images using vision transformer," *International Journal on Informatics Visualization*, vol. 8, no. 1, pp. 18–25, 2024.

- [8] M. Kalkatawi and U. Saeed, "Ethnicity classification based on facial images using deep learning approach," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 2, pp. 217–226, 2024.
- [9] R. A. Putri, L. Anifah, R. E. Putra, Y. Yamasari, and R. A. Akbar, "Dual vision transformer integration for race and gender recognition based on facial images," in *Proceedings of the 2025 8th International Conference on Vocational Education and Electrical Engineering (ICVEE)*, 2025, pp. 258–264.
- [10] R. A. Putri, R. E. Putra, and Y. Yamasari, "MD-ViT: Multidomain vision transformer fusion for fair demographic attribute recognition," *Journal of Information Engineering and Educational Technology*, vol. 9, no. 2, pp. 64–79, 2025.
- [11] I. Hupont and C. Fernández, "DemogPairs: Quantifying the impact of demographic imbalance in deep face recognition," in *Proceedings of the 2019 14th IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2019, pp. 1–7.



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